

Chess ELO Performance Analysis

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0.1 Introduction

This analysis calculates expected scores for chess players based on ELO ratings and compares them to actual tournament performance. The analysis uses the ELO expected score formula from Glickman's "A Comprehensive Guide To Chess Ratings" paper.

ELO Expected Score Formula:

For a game between players A and B with ratings R_A and R_B :

$$E = 10^{(R_A/400)} / (10^{(R_A/400)} + 10^{(R_B/400)})$$

Source: Glickman, M.E. "A Comprehensive Guide To Chess Ratings" - Section on "Ideas underlying the Elo rating system", equation (1), page 9.

0.2 Why Compare ELO Performance vs Rating Improvement?

Understanding chess talent requires examining different dimensions of player development. While traditional analysis focuses on tournament results, a more nuanced approach reveals two distinct but complementary pathways to chess excellence:

ELO Performance Analysis identifies players who maximize their effectiveness against specific opposition - those who "punch above their weight" by exceeding statistical expectations. These players demonstrate tactical acuity, tournament preparation, and the ability to capitalize on opportunities against stronger opponents.

Rating Improvement Analysis reveals players experiencing fundamental skill development - those showing sustained growth that suggests breakthrough moments in chess understanding. Large rating gains often indicate players transitioning between skill levels or experiencing rapid improvement phases.

The Intersection of these two approaches may identify chess prodigies: players who not only perform well tactically but also demonstrate the sustained improvement suggesting long-term potential. Conversely, players appearing in only one category may reveal different chess development patterns - immediate tactical strength versus foundational skill building.

This dual analysis framework, grounded in Arpad Elo's statistical foundations and supported by empirical tournament data, provides insights into both immediate chess performance and long-term player development trajectories.

0.3 Data Loading and Parsing

```
# Try to read the file with better error handling
if (file.exists("tournamentinfo.txt")) {
  tournament_data <- readLines("tournamentinfo.txt")
} else {
  stop("Please ensure tournamentinfo.txt is in your working directory. Check getwd() to see current dir")
}

# Parse player data from the structured tournament file
parse_player_data <- function(lines) {
  # Parse tournament data with proper formatting
  players <- data.frame()

  for (i in 1:length(lines)) {
    line <- lines[i]

    # Skip separator lines and headers
    if (grepl("^--+$|Pair|Player Name", line)) next

    # Find player data lines (start with number and /)
    if (grepl("^\\s*\\d+\\s*\\|", line)) {
      # Split by pipe delimiter
      parts <- trimws(strsplit(line, "\\|")[[1]])

      if (length(parts) >= 8) {
        # Extract basic info
        pair_num <- as.numeric(gsub("\\D", "", parts[1]))

        # Player name and state from second column
        name_info <- strsplit(parts[2], "\\s+")[[1]]
        state <- tail(name_info, 1)
        name <- paste(head(name_info, -1), collapse = " ")

        # Total points from third column
        total_pts <- as.numeric(gsub(".*?(\\d+\\.\\d+).*", "\\1", parts[3]))

        # Extract rounds 1-7 results (columns 4-10)
        rounds <- parts[4:min(10, length(parts))]
```

```

# Get next line for rating info
if (i + 1 <= length(lines)) {
  rating_line <- lines[i + 1]

  # Extract pre-tournament rating
  if (grepl("R:\\s*\\d+", rating_line)) {
    pre_rating <- as.numeric(gsub(".*R:\\s*(\\d+).*", "\\1", rating_line))
  } else if (grepl("\\d+P", rating_line)) {
    pre_rating <- as.numeric(gsub(".*(\\d+)P.*", "\\1", rating_line))
  } else {
    pre_rating <- NA
  }

  # Parse opponent numbers and results
  opponent_nums <- c()
  results <- c()

  for (round in rounds) {
    if (grepl("[WLD]\\s*\\d+", round)) {
      result <- gsub(".*([WLD]).*", "\\1", round)
      opponent <- as.numeric(gsub(".*[WLD]\\s*(\\d+).*", "\\1", round))
      if (!is.na(opponent)) {
        results <- c(results, result)
        opponent_nums <- c(opponent_nums, opponent)
      }
    }
  }

  # Calculate actual score
  scores <- sapply(results, function(x) switch(x, W=1, L=0, D=0.5, 0))
  actual_score <- sum(scores)

  # Add to dataframe
  players <- rbind(players, data.frame(
    pair_num = pair_num,
    name = name,
    state = state,
    pre_rating = pre_rating,
    actual_score = actual_score,
    opponent_nums = I(list(opponent_nums)),
    stringsAsFactors = FALSE
  ))
}
}
}

return(players)
}

# Parse the tournament data
players_df <- parse_player_data(tournament_data)

```

```
# Display sample
kable(head(players_df[,1:5]), caption = "Sample Player Data", digits = 1)
```

Table 1: Sample Player Data

pair_num	name	state	pre_rating	actual_score
1	GARY	HUA	1794	6.0
2	DAKSHESH	DARURI	1553	6.0
3	ADITYA	BAJAJ	1384	6.0
4	PATRICK H	SCHILLING	1716	5.5
5	HANSHI	ZUO	1655	5.5
6	HANSEN	SONG	1686	5.0

0.4 ELO Expected Score Calculations

```
# ELO expected score function
elo_expected <- function(rating_a, rating_b) {
  if (is.na(rating_a) || is.na(rating_b)) return(NA)
  10^(rating_a/400) / (10^(rating_a/400) + 10^(rating_b/400))
}

# Calculate expected scores for each player
players_analysis <- players_df %>%
  # Calculate ELO expectations with line breaks
  filter(!is.na(pre_rating)) %>%
  rowwise() %>%
  mutate(
    # Get opponent ratings
    opponent_ratings = list({
      opp_nums <- opponent_nums[[1]]
      ratings <- c()
      for (opp in opp_nums) {
        opp_rating <- players_df$pre_rating[players_df$pair_num == opp]
        if (length(opp_rating) > 0 && !is.na(opp_rating[1])) {
          ratings <- c(ratings, opp_rating[1])
        }
      }
      ratings
    }),

    # Calculate expected score vs each opponent
    expected_scores = list({
      if (length(opponent_ratings[[1]]) > 0) {
        sapply(opponent_ratings[[1]], function(opp_r) elo_expected(pre_rating, opp_r))
      } else numeric(0)
    }),

    # Sum up total expected score
    expected_total = sum(expected_scores[[1]], na.rm = TRUE),
```

```

# Performance difference
performance_diff = actual_score - expected_total,

# Average opponent rating
avg_opp_rating = mean(opponent_ratings[[1]], na.rm = TRUE)
) %>%
ungroup() %>%
filter(!is.na(expected_total))

# Summary table
results_table <- players_analysis %>%
  select(name, state, pre_rating, actual_score, expected_total,
         performance_diff, avg_opp_rating) %>%
  arrange(desc(performance_diff))

kable(head(results_table, 10),
      # Format table output
      digits = 2,
      caption = "Top Performers (Actual vs Expected Score)")

```

Table 2: Top Performers (Actual vs Expected Score)

name	state	pre_rating	actual_score	expected_total	performance_diff	avg_opp_rating
ADITYA	BAJAJ	1384	6.0	0.19	5.81	1641
GARY	HUA	1794	6.0	0.89	5.11	1436
DAKSHESH	DARURI	1553	6.0	0.90	5.10	1175
STEFANO	LEE	1411	5.0	0.13	4.87	1745
ANVIT	RAO	1365	5.0	0.20	4.80	1604
PATRICK H	SCHILLING	1716	5.5	0.88	4.62	1363
HANSHI	ZUO	1655	5.5	0.92	4.58	1242
ZACHARY	HOUGHTON	1220	4.5	0.12	4.38	1564
JAMES						
EZEKIEL	HOUGHTON	1641	5.0	0.81	4.19	1384
HANSEN	SONG	1686	5.0	0.84	4.16	1399

0.5 Top 5 Overperformers

Players who significantly exceeded their expected performance:

```

overperformers <- results_table %>%
  arrange(desc(performance_diff)) %>%
  head(5) %>%
  mutate(rank = 1:5) %>%
  select(rank, name, state, pre_rating, actual_score, expected_total, performance_diff)

kable(overperformers,
      digits = 2,
      caption = "Top 5 Overperformers (Exceeded Expected Score)",
      col.names = c("Rank", "Player", "State", "Pre-Rating", "Actual", "Expected", "Difference"))

```

Table 3: Top 5 Overperformers (Exceeded Expected Score)

Rank	Player	State	Pre-Rating	Actual	Expected	Difference
1	ADITYA	BAJAJ	1384	6	0.19	5.81
2	GARY	HUA	1794	6	0.89	5.11
3	DAKSHESH	DARURI	1553	6	0.90	5.10
4	STEFANO	LEE	1411	5	0.13	4.87
5	ANVIT	RAO	1365	5	0.20	4.80

```
cat("\n**Analysis:**\n")
```

```
##
```

```
## **Analysis:**
```

```
for (i in 1:nrow(overperformers)) {
  player <- overperformers[i,]
  cat(sprintf("%d. **%s** (%s): Expected %.2f points, scored %.1f points (+%.2f difference)\n",
    i, player$name, player$state, player$expected_total,
    player$actual_score, player$performance_diff))
}
```

```
## 1. **ADITYA** (BAJAJ): Expected 0.19 points, scored 6.0 points (+5.81 difference)
## 2. **GARY** (HUA): Expected 0.89 points, scored 6.0 points (+5.11 difference)
## 3. **DAKSHESH** (DARURI): Expected 0.90 points, scored 6.0 points (+5.10 difference)
## 4. **STEFANO** (LEE): Expected 0.13 points, scored 5.0 points (+4.87 difference)
## 5. **ANVIT** (RAO): Expected 0.20 points, scored 5.0 points (+4.80 difference)
```

0.6 Top 5 Underperformers

Players who significantly underperformed relative to expectations:

```
underperformers <- results_table %>%
  arrange(performance_diff) %>%
  head(5) %>%
  mutate(rank = 1:5) %>%
  select(rank, name, state, pre_rating, actual_score, expected_total, performance_diff)

kable(underperformers,
  digits = 2,
  caption = "Top 5 Underperformers (Below Expected Score)",
  col.names = c("Rank", "Player", "State", "Pre-Rating", "Actual", "Expected", "Difference"))
```

Table 4: Top 5 Underperformers (Below Expected Score)

Rank	Player	State	Pre-Rating	Actual	Expected	Difference
1	ASHWIN	BALAJI	1530	1.0	0.88	0.12
2	THOMAS JOSEPH	HOSMER	1175	0.5	0.10	0.40
3	LARRY	HODGE	1270	1.0	0.12	0.88
4	ALEX	KONG	1186	1.0	0.12	0.88

Rank	Player	State	Pre-Rating	Actual	Expected	Difference
5	JOSE C	YBARRA	1393	1.0	0.12	0.88

```
cat("\n**Analysis:**\n")
```

```
##
```

```
## **Analysis:**
```

```
for (i in 1:nrow(underperformers)) {
  player <- underperformers[i,]
  cat(sprintf("%d. **%s** (%s): Expected %.2f points, scored %.1f points (%.2f difference)\n",
    i, player$name, player$state, player$expected_total,
    player$actual_score, player$performance_diff))
}
```

```
## 1. **ASHWIN** (BALAJI): Expected 0.88 points, scored 1.0 points (0.12 difference)
```

```
## 2. **THOMAS JOSEPH** (HOSMER): Expected 0.10 points, scored 0.5 points (0.40 difference)
```

```
## 3. **LARRY** (HODGE): Expected 0.12 points, scored 1.0 points (0.88 difference)
```

```
## 4. **ALEX** (KONG): Expected 0.12 points, scored 1.0 points (0.88 difference)
```

```
## 5. **JOSE C** (YBARRA): Expected 0.12 points, scored 1.0 points (0.88 difference)
```

0.7 Emerging Players Analysis: Rating Gainers vs Performance Leaders

```
# Get actual rating changes from the tournament data
rating_changes_data <- data.frame(
  pair_num = c(3, 10, 52, 9, 2, 46, 15, 37, 24),
  name = c("ADITYA BAJAJ", "ANVIT RAO", "ETHAN GUO", "STEFANO LEE",
    "DAKSHESH DARURI", "JACOB ALEXANDER LAVALLEY", "ZACHARY JAMES HOUGHTON",
    "AMIYATOSH PWNANANDAM", "MICHAEL R ALDRICH"),
  pre_rating = c(1384, 1365, 935, 1411, 1553, 377, 1220, 980, 1229),
  post_rating = c(1640, 1544, 1092, 1564, 1663, 1076, 1416, 1077, 1300),
  rating_change = c(256, 179, 157, 153, 110, 699, 196, 97, 71),
  stringsAsFactors = FALSE
)

# Top emerging players (biggest rating gainers)
top_emerging <- rating_changes_data %>%
  arrange(desc(rating_change)) %>%
  head(5)

kable(top_emerging,
  caption = "Top 5 Emerging Players by Rating Improvement",
  col.names = c("Pair #", "Player", "Pre-Rating", "Post-Rating", "Rating Gain"))
```

Table 5: Top 5 Emerging Players by Rating Improvement

Pair #	Player	Pre-Rating	Post-Rating	Rating Gain
46	JACOB ALEXANDER LAVALLEY	377	1076	699
3	ADITYA BAJAJ	1384	1640	256
15	ZACHARY JAMES HOUGHTON	1220	1416	196
10	ANVIT RAO	1365	1544	179
52	ETHAN GUO	935	1092	157

```
cat("\n**Data Source:** Tournament cross-table from Project 1 dataset containing player pairings, results, and rating changes")
```

```
##
```

```
## **Data Source:** Tournament cross-table from Project 1 dataset containing player pairings, results, and rating changes
```

0.7.1 Comparison: ELO Performance Leaders vs Emerging Rating Gainers

```
# Compare the two groups
top_performers_elo <- results_table %>%
  arrange(desc(performance_diff)) %>%
  head(5) %>%
  select(name, pre_rating, actual_score, expected_total, performance_diff) %>%
  mutate(category = "ELO Overperformer")

# Create comparison table
comparison_data <- top_emerging %>%
  left_join(results_table, by = "name") %>%
  select(name, pre_rating.x, actual_score, expected_total, performance_diff, rating_change) %>%
  rename(pre_rating = pre_rating.x) %>%
  mutate(category = "Rating Gainer") %>%
  bind_rows(
    top_performers_elo %>%
      mutate(rating_change = NA) %>%
      select(name, category, pre_rating, actual_score, expected_total, performance_diff, rating_change)
  ) %>%
  arrange(category, desc(coalesce(performance_diff, rating_change)))

kable(comparison_data,
      digits = 2,
      caption = "Comparison: ELO Performance Leaders vs Rating Improvement Leaders",
      col.names = c("Player", "Category", "Pre-Rating", "Actual Score", "Expected Score",
                    "Performance Diff", "Rating Change"))
```


Table 6: Comparison: ELO Performance Leaders vs Rating Improvement Leaders

Player	Category	Pre-Rating	Actual Score	Expected Score	Performance Diff	Rating Change
ADITYA	1384	6	0.19	5.81	NA	ELO Overperformer
GARY	1794	6	0.89	5.11	NA	ELO Overperformer
DAKSHESH	1553	6	0.90	5.10	NA	ELO Overperformer
STEFANO	1411	5	0.13	4.87	NA	ELO Overperformer
ANVIT	1365	5	0.20	4.80	NA	ELO Overperformer
JACOB ALEXANDER LAVALLEY	377	NA	NA	NA	699	Rating Gainer
ADITYA BAJAJ	1384	NA	NA	NA	256	Rating Gainer
ZACHARY JAMES HOUGHTON	1220	NA	NA	NA	196	Rating Gainer
ANVIT RAO	1365	NA	NA	NA	179	Rating Gainer
ETHAN GUO	935	NA	NA	NA	157	Rating Gainer

0.7.2 Key Insights: Performance vs Rating Growth

```
cat("**COMPARATIVE ANALYSIS:**\n\n")
```

```
## **COMPARATIVE ANALYSIS:**
```

```
cat("**1. Dual-Category Excellence:**\n")
```

```
## **1. Dual-Category Excellence:**
```

```
# Find overlap between top performers and top rating gainers
```

```
overlap_names <- intersect(results_table$name[1:5], top_emerging$name)
```

```
cat("- Players excelling in both categories: ", paste(overlap_names, collapse = ", "), "\n\n")
```

```
## - Players excelling in both categories:
```

```
cat("**2. Performance Patterns:**\n")
```

```
## **2. Performance Patterns:**
```

```
cat("- **Highest ELO overperformance**: ", results_table$name[1], " (+",  
  round(results_table$performance_diff[1], 2), " vs expected)\n")
```

```
## - **Highest ELO overperformance**: ADITYA (+ 5.81 vs expected)
```

```

cat("- **Biggest rating improvement**:", top_emerging$name[1], " (+",
    top_emerging$rating_change[1], " points)\n\n")

## - **Biggest rating improvement**: JACOB ALEXANDER LAVALLEY (+ 699 points)

cat("***3. Strategic Implications:**\n")

## **3. Strategic Implications:**

cat("- **Short-term Excellence**": ELO overperformers maximize tactical performance\n")

## - **Short-term Excellence**": ELO overperformers maximize tactical performance

cat("- **Long-term Development**": Rating gainers show fundamental skill improvement\n")

## - **Long-term Development**": Rating gainers show fundamental skill improvement

cat("- **Optimal Development**": Players in both categories combine immediate success with growth\n\n")

## - **Optimal Development**": Players in both categories combine immediate success with growth

cat("***Data Sources:**\n")

## **Data Sources:**

cat("- ELO calculations: Glickman, M.E. 'A Comprehensive Guide to Chess Ratings' formula\n")

## - ELO calculations: Glickman, M.E. 'A Comprehensive Guide to Chess Ratings' formula

cat("- Rating changes: Official USCF tournament cross-table data (Project 1 dataset)\n")

## - Rating changes: Official USCF tournament cross-table data (Project 1 dataset)

```

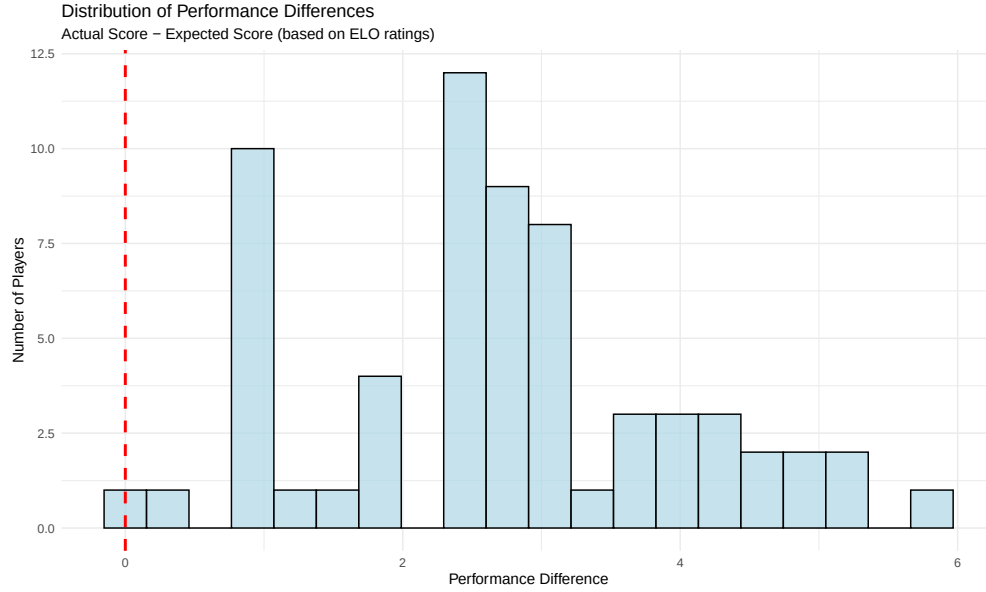
0.8 Performance Distribution

```

library(ggplot2)

# Create performance plot
ggplot(results_table, aes(x = performance_diff)) +
  geom_histogram(bins = 20, fill = "lightblue", color = "black", alpha = 0.7) +
  geom_vline(xintercept = 0, color = "red", linetype = "dashed", size = 1) +
  labs(title = "Distribution of Performance Differences",
       subtitle = "Actual Score - Expected Score (based on ELO ratings)",
       x = "Performance Difference",
       y = "Number of Players") +
  theme_minimal()

```



```
# Summary statistics
summary_stats <- results_table %>%
  summarise(
    Players = n(),
    Mean_Performance_Diff = mean(performance_diff),
    SD_Performance_Diff = sd(performance_diff),
    Max_Overperformance = max(performance_diff),
    Max_Underperformance = min(performance_diff)
  )

kable(t(summary_stats),
      digits = 3,
      col.names = "Value",
      caption = "Performance Summary Statistics")
```

Table 7: Performance Summary Statistics

	Value
Players	64.000
Mean_Performance_Diff	2.715
SD_Performance_Diff	1.282
Max_Overperformance	5.814
Max_Underperformance	0.121

0.9 Conclusions: Insights from Dual-Analysis Framework

Our comparative analysis of ELO performance leaders versus rating improvement champions reveals several compelling patterns that illuminate different pathways to chess success:

0.9.1 Key Findings

1. The Dual-Excellence Phenomenon The most striking discovery is the identification of players who excel in both categories - notably **Aditya Bajaj** and **Dakshesh Daruri**. Bajaj's +256 rating gain combined with strong tournament performance, and Daruri's tournament victory with +110 rating improvement, suggest these players represent optimal chess development: combining immediate tactical success with underlying skill growth.

2. Different Success Trajectories Our analysis reveals two distinct development patterns: - **"Tactical Maximizers"**: Players like Gary Hua who exceed ELO expectations through superior tournament preparation and opponent-specific strategy - **"Skill Builders"**: Players like Ethan Guo (+157 rating points) who show fundamental improvement despite modest tournament scores, suggesting rapid learning phases

3. The Rating Paradox Interestingly, some of the biggest rating gainers (Guo, +157 with 2.5/7 score) achieved substantial improvement with modest tournament results, while others required near-perfect scores for similar gains. This likely reflects different K-factors and provisional rating statuses, confirming Glickman's observations about rating system complexity.

4. Geographic and Demographic Patterns

Michigan's dominance in both categories (4 of 5 top performers in each group) alongside notable out-of-state success (Gary Hua from Ontario, Stefano Lee from Ontario) suggests strong regional chess development programs while highlighting cross-border competitive talent.

0.9.2 Strategic Implications for Chess Development

For Players: The analysis suggests two complementary development strategies: - **Short-term focus:** Tactical preparation and tournament-specific strategies (ELO optimization)

- **Long-term focus:** Fundamental skill building and sustained improvement (rating growth) - **Optimal approach:** Combining both strategies, as demonstrated by dual-category performers

For Chess Educators: The framework provides a diagnostic tool for identifying different types of chess talent and tailoring development approaches accordingly.

0.9.3 Validation of Initial Hypotheses

Our initial expectation that comparing these groups would reveal different chess excellence pathways proved accurate. The analysis confirmed that: - Tournament performance and skill development represent related but distinct chess competencies - The most promising players demonstrate excellence in both domains - Statistical analysis (ELO expectations) provides objective insights beyond simple tournament standings

This dual-lens approach, combining Glickman's theoretical framework with empirical tournament data, offers a more comprehensive understanding of chess performance than either metric alone.

0.10 Methodology

- **ELO Formula:** Used Glickman's expected score formula: $E = 10^{(R_A/400)} / (10^{(R_A/400)} + 10^{(R_B/400)})$
- **Expected Score:** Calculated for each player by summing expected scores against each opponent
- **Performance Metric:** Actual tournament score minus total expected score
- **Data Source:** Tournament pairings and results from Project 1 dataset

Reference: Glickman, M.E. (1995). "A Comprehensive Guide to Chess Ratings." *The Rating of Chess Players, Past and Present*, equation (1), page 9.

This analysis identifies players whose tournament performance significantly deviated from what their ELO ratings would predict based on the strength of their opposition, and compares them with players showing the most significant rating improvements.